A Weighted-Sum Approach for Multi-Objective Optimization of Strong Barrier Coverage in IoT Heterogeneous Wireless Sensor Networks

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Abstract. Strong barrier coverage (SBC) is a critical requirement for IoT heterogeneous wireless sensor networks (IoT-HeWSNs), particularly in applications such as border surveillance, environmental monitoring, and intrusion detection. The achievement of optimal SBC in IoT-HeWSNs presents significant challenges due to the diversity of sensor capabilities, the conflicting nature of multiple objectives, and constraints on energy efficiency and deployment costs. Traditional methods often address the SBC problem using single-objective optimization, focusing on energy consumption, sensor deployment costs, or isolation coverage efficiency. To overcome these limitations, we formulate the multi-objective optimization problem for strong barrier coverage (MOO-SBC) and propose advanced optimization methods that simultaneously optimize sensor count, energy usage, and network coverage. Our approach, termed WS-SBC, employs state-of-the-art evolutionary algorithms to achieve superior trade-offs among these objectives. Extensive simulations demonstrate significant reductions in energy consumption and sensor count, along with ensured strong barrier coverage. The proposed method provides an efficient solution that balances energy consumption and the number of sensors. The achieved results improve IoT-HeWSN performance, offering practical applicability in real-world scenarios that require robust and cost-effective coverage solutions.

Keywords: Barrier coverage · Multi-objective optimization of strong barrier coverage · Heterogeneous wireless sensor networks· Weighted-Sum approach.

1 Introduction

Wireless sensor networks (WSNs) are composed of numerous small, low-cost, energy-efficient sensor nodes that operate wirelessly over short distances [16]. These sensor nodes are typically deployed randomly across the target area and are widely used for monitoring and surveillance tasks. Depending on the application, WSNs must optimize various performance metrics, such as energy efficiency and network lifetime, which are crucial because these nodes are usually

battery-powered, making replacements challenging [10,12]. The proliferation of IoT heterogeneous wireless sensor networks (IoT-HeWSNs) has revolutionized applications such as border surveillance, environmental monitoring, and intrusion detection [8,17,9,10,1]. In these contexts, strong barrier coverage (SBC) plays a crucial role in ensuring reliable detection and tracking across designated areas [16]. Unlike traditional area coverage, strong barrier coverage ensures a fortified line of sensors that acts as a virtual fence [4,7]. However, achieving optimal SBC in IoT-HeWSNs is challenging due to the heterogeneous nature of sensor capabilities, varying energy constraints, and conflicting design objectives.

Traditional methods for addressing the SBC problem often rely on single-objective optimization, focusing on metrics such as energy efficiency, deployment costs, or coverage quality [20,10,5,7,4]. While these approaches offer insights into individual aspects of network performance, they fail to account for the inherent trade-offs between these competing objectives, leading to suboptimal solutions in real-world deployments. The complexity increases further in heterogeneous networks, where diverse sensor types must work in synergy to achieve efficient and effective barrier coverage.

To tackle these challenges, this paper presents a weighted-sum approach for multi-objective optimization of strong barrier coverage (MOO-SBC) in IoT-HeWSNs. The proposed method integrates advanced evolutionary algorithms to balance two key objectives: minimizing sensor deployment costs or sensor count, reducing energy consumption. By assigning weights to these objectives, our approach enables network designers to prioritize specific performance metrics while maintaining a holistic optimization framework. We have considered it critical and challenging to investigate the following problem: for a given surveillance sensing field, determine the locations and the transmit power levels of a limited number of sensors to be deployed, to minimize the number of active sensors and consumption energy, at the same time. The main contributions of this work are as follows:

- We formally define the MOO-SBC problem, incorporating the unique characteristics and constraints of IoT-HeWSNs.
- We propose a weighted-sum approach that utilizes state-of-the-art evolutionary algorithms to achieve optimal trade-offs among multiple objectives, known as SW-SBC. → WS - SBC
- Through extensive simulations, we validate the effectiveness of our approach, demonstrating significant improvements in network lifetime, cost efficiency, and robustness of barrier coverage compared to traditional methods.

The rest of this paper is organized as follows: Section 2 presents a survey of related literature. Preliminaries and problem formulation for the maximal exposure path are discussed in Section 3. Section 4 introduced the proposed methodology, while experimental results are described and analyzed in Section 5. Finally, Section 6 draws the conclusions of the paper.

2 Related Works

Barrier coverage (BC) in wireless sensor networks (WSNs) is a critical area of research, particularly for Internet of Things (IoT) applications that require robust security and reliable sensor network operation. Strong barrier coverage (SBC) in WSNs ensures that intrusions are detected along a defined line of defense, creating a continuous barrier of detection capabilities. BC problems in WSNs are typically classified into two categories: weak and strong barrier coverage.

Weak barrier coverage involves identifying penetration paths through the sensing field and has garnered significant attention from research communities [2,15,19,6]. These studies primarily investigate the minimal exposure path problem in WSNs under various assumptions. On the other hand, SBC has gained increasing attention in applications requiring high reliability, where every point along the barrier line must be monitored by at least one sensor. The authors in [7] introduced the concept of SBC, emphasizing optimal sensor deployment to maximize coverage probability. Recent advancements in this field have incorporated heterogeneous sensors in IoT networks, leveraging the advantages of node heterogeneity, such as enhanced network lifetime and reliability [9,13,3,14,4]. For instance, [9] provides a survey on coverage optimization using data fusion HeWSNs. In [13], the authors proposed a novel Delay-Aware Energy-Efficient Reliable Routing technique for data transmission in heterogeneous sensor environments, introducing a restricted search space to guarantee timely delivery of delay-sensitive data. Studies in [3,14,4] focus on finding the best-case coverage penetration path under various constraints. Although these works present unique challenges related to variations in sensing range, energy consumption, and processing capabilities among nodes, they often address specific problems. However, it is important to note that the majority of these studies primarily focus on single-objective optimization, leaving room for further exploration of multi-objective approaches.

Multi-objective optimization (MOO) has been applied in WSNs to balance conflicting objectives such as energy consumption, coverage, and network lifetime. Researchers have developed several evolutionary algorithms, such as the non-dominated sorting genetic algorithm II and particle swarm optimization, the weighted-sum Method, among others to address these issues. The researchers in [11,18] reviewed multiple MOO methods used in WSNs, highlighting the effectiveness of weighted-sum and ϵ-constraint methods for specific cases where objectives could be adequately balanced by adjusting priority weights. The weighted-sum method has proven effective for MOO in cases where objectives align along convex Pareto fronts, as it allows for a straightforward balance among key objectives by assigning weights to each criterion. The authors in [21] define three objectives to minimize: (1) total power consumption while ensuring full coverage, (2) the number of active sensors to enhance reliability, and (3) the maximum sensing range of active sensor nodes to promote fairness. They propose a multi-objective optimization framework based on the multi-objective evolutionary algorithm. However, the third objective is questionable, as a larger sensing range

typically leads to higher energy consumption. Additionally, the sensing coverage model used in this work is a binary model, which is an idealized representation. Therefore, it is necessary to enhance the problem formulation by incorporating a more realistic sensing model.

After thoroughly studying related research on multi-objective optimization for the strong barrier coverage problem in HWSNs, the findings reveal that there are still limitations and issues that need to be addressed. This is necessary to solve the problem under more realistic assumptions and make it easier to apply in practical applications.

3 Preliminaries and Problem Formulation

3.1 Preliminaries

Definition 1 (Sensing coverage model). *This paper focuses on the probabilistic sensing model or truncate attenuated disk coverage model is given by following equation 1*

$$\mathcal{C}(S_i, O) = \begin{cases} 1 & \text{if } d(S_i, O) \leq R_s - R_U \\ e^{-\alpha\theta} & \text{if } R_s - R_U < d(S_i, O) \leq R_s \\ 0 & \text{if } d(S_i, O) > R_s \end{cases} \tag{1}$$

where, $d(s_i, O)$ represents the Euclidean distance between sensor s_i at a location (x_i, y_i) and a point or target O located at position (x, y), i.e. $d(s_i, O) = \sqrt{(x_i - x)^2 + (y_i - y)^2}$, the uncertainty detection of the sensor nodes is defined by R_U ($R_U \leq R_s$), and α, θ, are parameters related to the detection probability at distances less than or equal to R_s. Here, $\theta = \beta \cdot (d(O, s_i) - (R_s - R_U))$. Equation 1 indicates that a point is considered covered if it lies within a distance of $(R_s - R_U)$ from a sensor node. Coverage decreases exponentially as the distance between a point and the sensor node increases if the point is within the range $(R_s - R_U, R_s)$. Points beyond the distance R_s are regarded as uncovered. Figure 1a illustrates for the truncate attenuated disk coverage model.

Definition 2 (Sensing intensity model). *In a network, it is possible that a point might be covered by multiple (N) sensor nodes, $S = \{S_1, S_2, .., S_N\}$, in which case the point is known as N-covered.*

Intensity

$$\mathcal{I}(S, O) = \sum_{i=1}^{N} \mathcal{C}(S_i, O) \tag{2}$$

Definition 3 (Mathematical Model for Energy Consumption). *The total energy consumption, E_{total}, is derived as the sum of two integrals. This approach divides energy consumption into two parts to account for both core and attenuated energy requirements within the sensor range:*

$$E_{total} = \int_0^{(k-1)\cdot R_U} R\, dR + \int_{(k-1)\cdot R_U}^{k\cdot R_U} e^{-\alpha\beta[R-(k-1)\cdot R_U]} dR \tag{3}$$

The first integral represents the baseline energy required for the primary sensing range, where energy is consumed proportionally to R. The second integral accounts for the additional energy required as the sensor signal extends beyond the primary range. Here, the energy decreases exponentially, capturing the natural attenuation of signal strength with increasing distance. This combined formula provides a comprehensive model for total energy consumption, balancing constant core energy needs with distance-dependent signal attenuation.

$$\mathtt{E}_{total} = \sum_{s_i \in S} \frac{1}{2}[(k-1) \cdot R_U]^2 + \frac{(k-1) \cdot R_U \cdot (1-\omega)}{\beta} + \frac{1-\omega \cdot (1+\beta \cdot R_U)}{\beta^2} \quad (4)$$

with $\omega = \exp(-\beta \cdot R_U)$

Definition 4 (Strong barrier coverage model). *A strong sensor barrier is established by interconnecting a cluster of sensors throughout the entire surveillance region, serving as a "virtual fence" to identify any intruders attempting to cross.*

The concept of Strong barrier coverage implies that every point on a line segment L is covered by a set $\mathcal{S} = (S_1, S_2, ..S_N)$ of wireless sensors that have adjustable coverage ranges. Each sensor S_i, i=(1..N), has a set $R(S_i)$ of possible coverage radii. For any $r \in R(S_i)$, the coverage range of sensor s_i spans an interval $\mathcal{C}(S_i, r)$ of length $2r$, centered at S_i, with an associated cost $c(S_i, r)$. This cost proportional to $r^{\mathcal{K}}$, where $\mathcal{K}$ represents the path-loss exponent, a constant between 2 and 5 that depends on the wireless environment [1]. Figure 1b demonstrates strong barrier coverage in HWSN.

The cost can be further modified based on the desired quality of monitoring. For instance, the mean power assignment proposed by [4] reduces the exponent $\mathcal{K}$ by half, resulting in a range $[1, 2.5]$. For simplicity, we assume that for each sensor S_i, $0 \in R(S_i)$ and $c(S_i, 0) = 0$, representing no coverage cost when the sensor is inactive. The objective is to choose a coverage radius $r(S_i) \in R(S_i)$ for each sensor $S_i \in \mathcal{S}$ such that

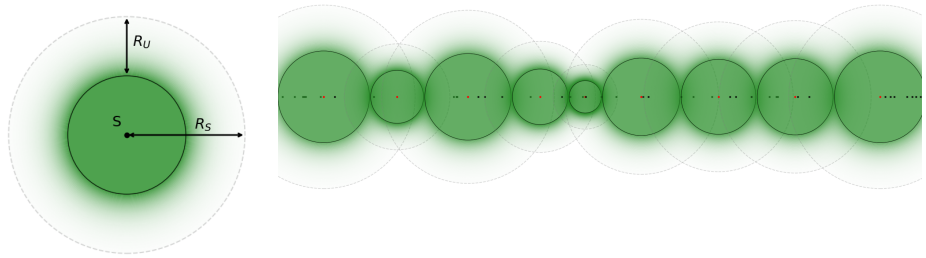
$$L \subseteq \bigcup_{S_i=1}^{N} C(S_i, r(S_i)) \quad (5)$$

3.2 Problem Formulation

Let us assume that N heterogeneous static sensors $S = \{S_1, S_2, ..., S_N\}$ are randomly deployed in the ROI of dimensions $L \times H$. These sensors can be either activated or deactivated. Each sensor nodes has an adjustable sensing range R_i, and the sensing coverage model is probabilistic. After deployment, each sensor s_i has a unique position (x_i, y_i) and a specific sensing radius R_i. Furthermore, the sensing range R_i is significantly smaller than the length and width of the ROI to be covered, rendering the boundary effect negligible and thus ignorable. With this radius, the coverage range of sensor S_i can be represented as the interval

$C(S_i, R_i)$ of length $2R_i$ centered at x_i. Additionally, each sensor is associated with a cost $c(S_i, R_i)$, this cost represents the energy consumption during operation. The objective is to select appropriate coverage radii R_i to optimize both deployment cost and energy consumption while ensuring strong barrier coverage across the ROI.

$\int_0^{R_s} 2R\pi \, dr$



(a) Truncate attenuated disk coverage model

(b) Strong barrier coverage

Fig. 1: Illustration of sensing coverage model and strong barrier coverage model under truncate multilevel attenuated disk

The objective is to choose a coverage radius R_i for each sensor s_i such that

$$L \subseteq \bigcup_{s_i \in S} C(s_i, r)$$

and the total cost of the deployment $\sum_{s_i \in S} c(s_i, r)$ is minimized. This optimization problem is referred to as min-cost-linear-coverage.

Input:

- N: the number of deployed sensors.
- $X = \{x_i\}$: x coordinates of N sensors.
- L: barrier length
- γ: detection threshold
- α, β: attenuation coefficient of the sensor
- k: R_s to R_U ratio R_s / R_U

Output:

- A set of solutions to activate/deactivate sensors along with their radii R_s

Decision variable:

– Using status of sensor s_i

$$\mu_i = \begin{cases} 1 & \text{if} \quad \text{sensor } s_i \text{ is active} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

– The width of the sensor uncertainty range, R_U

$$0 \leq R_U \leq R_{U_{max}} \tag{7}$$

Objective:

– Reliability: Minimizing the number of active sensors:

$$f_1 = \sum_{s_i \in S} \mu_i \tag{8}$$

– Power: Minimizing the total power consumption:

$$f_2 = \mathtt{E}_{total} \tag{9}$$

Constraints: The sensing range R_i is much smaller than the length, the width of ROI to be covered, so that the boundary effect is negligible and hence can be ignored.

4 Proposed Algorithm

The linear weighted-sum method transforms multiple performance metrics into a single-objective function by assigning a weight to each metric (or individual objective) and then multiplying each by its respective weight. Because the metrics vary in properties and may use different units, normalization is necessary before applying the weighted-sum method to ensure balanced performance trade-offs. After normalization, each metric is given a specific weight to form an evaluation function. The final optimal solution is derived from Pareto-optimal solutions by solving several single-objective problems, each based on a different weight vector. It can be shown that the optimal solution to each single-objective problem is also a Pareto-optimal solution to the initial multi-objective problem. The pseudocode for the proposed weighted-sum algorithm, WS-SBC, is provided in Algorithm 1 below.

4.1 Individual Presentation

Each individual is encoded as a binary string, where each bit represents a sensor's state: 1 for active, 0 for inactive. The string length matches the total number of sensors, ensuring each individual corresponds to a unique activation setup.

Algorithm 1 SW-SBC

Input: X: x-coordinates of the set of sensors
Output: P: Set of final solutions
Initialize population P_0, a set of weight vectors $\{\lambda_i\}$, ideal point z^*, and nadir point z_{nad}
for generation = 1 to $max_generation$ **do**
  for each subproblem i in the population **do**
    Select mating partners using neighborhood structure
    Generate offspring y through crossover and mutation operations
    Evaluate the objective functions for y using the **Evaluate** algorithm
    if $f(y|\lambda_i) < f(x_i|\lambda_i)$ **then**
      Replace x_i with y in the population
      Update ideal point z^*
    end if
  end for
  Update nadir point z_{nad}
end for
return Final population P

4.2 Population Initialization

In the proposed algorithm, SW-SBC, we initialize T sets of lambda values, where T is the population size, and distribute them evenly across individuals. Each individual is initialized randomly to ensure no all-zero configurations.

4.3 Evolution Operators

Crossover: We use synchronous crossover, selecting genes for offspring based on a random value $u \in [0, 1]$: if $u < 0.5$, child 1 inherits the gene from the male parent; otherwise, from the female. Child 2 receives the opposite gene.
Mutation: A bit-flip mutation randomly flips m bits in each individual's chromosome.

4.4 Radius Formalization Algorithm

The Radius Formalize Algorithm is designed to determine the valid radius range for each sensor in a HeWSN deployment scenario. This step ensures that sensor coverage and energy efficiency are optimized within the constraints of the problem. The algorithm first checks whether the expanded radius of the current sensor can already reach the next sensor. If so, it returns a zero value, indicating no additional radius is needed. A search is conducted to find the minimum x-coordinate that satisfies a signal strength condition. The pseudocode for the Radius Formalization algorithm is represented in Algorithm 2 as follows:

Algorithm 2 Radius Formalization

Input: r_{u1}, x_1, x_2
Output: Valid value of r_{u2} or 0 if no valid value is found
Note: x_1 and r_{u1} represent the x-coordinate and radius of the current sensor, respectively, while x_2 and r_{u2} correspond to the next sensor to the right.

$r^*_{u1} \leftarrow r_{u1} \cdot (k-1)$ → R_s/R_w
if $x_1 + r^*_{u1} \geq x_2$ **then**
  return 0
end if
Define $term(x) \leftarrow x - x_1 - r^*_{u1}$ $\quad x - (x_1 + r_{u1})$
for $x \leftarrow (x_1 + r_{u1})$ to x_2 **do**
  if $\exp(\beta \cdot term(x)) \geq \gamma$ **then** gần x_1 nhất: $e^{\beta \cdot term(x)} \geq \gamma$
    $x_{1_certain} \leftarrow x$
    break → ?
  end if
end for
if $x_{1_certain} \geq x_2$ **then**
  return 0
end if
$x_{1_certain} \leftarrow \text{floor}(x_{1_certain})$ $\quad - r_{u2}\left(\frac{R_s}{R_w} - 1\right) =$ → phần phía trong $x_2 - x_1$ certain
$r_{u2_max} \leftarrow \frac{x_2 - x_{1_certain}}{k-1}$
$r_{u2_min} \leftarrow \frac{x_2 - x_{1_certain}}{k}$ → R_s =
for $r_{u2} \leftarrow \text{ceil}(r_{u2_min})$ to $\text{ceil}(r_{u2_max})$ **do**
  $r^*_{u2} \leftarrow (k-1) \cdot r_{u2}$
  $x_w \leftarrow \frac{x_1 + x_2 - (k-1)\cdot(r_{u1} - r_{u2})}{2}$ ▷ x_w is the midpoint where the signal received is the weakest between the two sensors
  $signal_w \leftarrow \exp(\beta \cdot (x_w - x_1 - r^*_{u1}))$
    $+ \exp(\beta \cdot (x_2 - x_w - r^*_{u2})) - \gamma$
  if $signal_w \geq 0$ **then**
    $valid_r_{u2} \leftarrow r_{u2}$
    break
  end if
end for
if $valid_r_{u2} \neq$ None **then**
  return $valid_r_{u2}$
end if
return 0

5 Numerical Results

5.1 Parameter settings

Experimental Data
The dataset was randomly initialized following a uniform distribution within the range from 1 to the barrier length (1000). Each dataset consisted of unique integer coordinate points selected randomly from this range. After generation, the points were sorted in ascending order to simulate ordered positions along a line. The experiments were conducted with datasets of varying sizes, specifically with N set to 100, 150, 200, and 250, representing the number of coordinate points in each scenario. For each value of N, experiments were performed on 10 instances, labeled as $N_0, N_1, \ldots, N_9$, resulting in a total of 40 instances to ensure the robustness of the results.
Experimental environment
The experiments were conducted on a computer equipped with an Intel processor running at 1 GHz and 8 GB of RAM, ensuring sufficient computational resources for testing. The implementation was carried out using Python. The environment was configured to handle multiple runs efficiently, ensuring reproducibility and reliability of the results.
Experimental parameters

Parameter	**Value**
Barrier length	1000
Detection threshold	0.5
Attenuation coefficient	1
R_S to R_U ratio	2
The number of running on each instance	50
The number of Generations	1000
The population size	32
The crossover rate	100%
The mutation rate	30%
The neighbor size	6

5.2 Experimental results

5.2.1. Evaluating performance of proposed algorithm through the hyper volume metric
The hypervolume metric is crucial in multi-objective optimization because it provides a comprehensive measure of both quality and diversity of the solutions found by the algorithm. Hypervolume captures how close the solutions generated by an algorithm are to the true Pareto front (the set of optimal trade-offs among objectives). A larger hypervolume indicates that solutions are closer to this optimal front, implying better performance. By considering the volume covered by all non-dominated solutions, the hypervolume metric also reflects the spread of solutions across different regions of the objective space. This diversity is essential for decision-makers who may want a variety of trade-offs among objectives. Furthermore, instead of evaluating each objective separately, the hypervolume metric synthesizes them into one value. This makes it easier to assess overall performance without separately analyzing each objective. Figure 2a and 2b shows the convergence of a multi-objective optimization algorithm over 1000 generations on the data instance 100 and 150 number of sensors, respectively. The

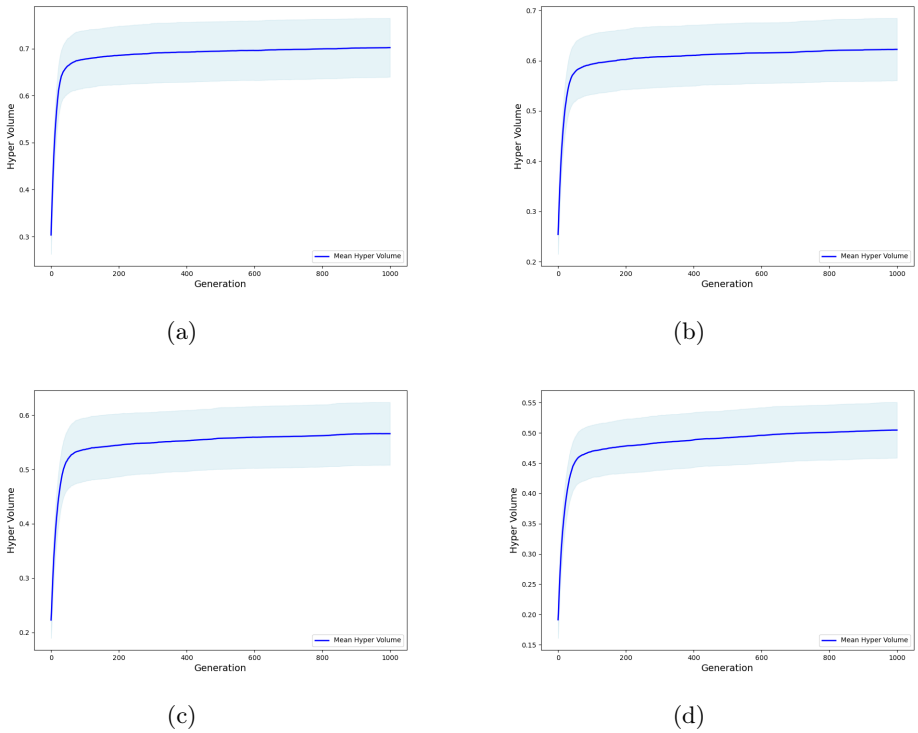


Fig. 2: Hypervolume results for different sensor configurations: (a) 100 sensors, (b) 150 sensors, (c) 200 sensors, and (d) 250 sensors.

hypervolume metric is used to measure the quality of the Pareto front, which represents the set of non-dominated solutions. Observationally, the hypervolume increases rapidly in the initial generations, indicating that the algorithm quickly explores the search space and finds good solutions. Furthermore, the hypervolume plateaus around generation 400, suggesting that the algorithm has converged to a near-optimal Pareto front. Besides, the shaded region represents the standard deviation of the hypervolume across multiple runs, indicating the variability in the algorithm's performance. In short, the algorithm is effective in finding a good Pareto front within a reasonable number of generations. The convergence rate is high, especially in the initial stages. In addition, the algorithm is relatively stable, with a small standard deviation in the hypervolume across different runs.

Figure 2c and 2d show the convergence of a multi-objective optimization algorithm for the two different sensor configurations: 200 sensors (a) and 250 sensors (b). The hypervolume metric, which measures the quality of the Pareto front, increases rapidly in the initial stages and then plateaus, indicating that the algorithm effectively explores the search space and converges to a near-optimal solution.

Additionally, we can observe that the algorithm with 250 sensors converges to a higher hypervolume than the algorithm with 200 sensors. This suggests that increasing

the number of sensors can improve the overall performance of the system. In addition, the convergence rate is similar for both configurations.This indicates that the algorithm is able to handle different numbers of sensors efficiently. These findings suggest that the multi-objective optimization algorithm is robust and effective in optimizing sensor networks with varying numbers of sensors.

In summary, the hypervolume metric is essential because it efficiently combines closeness to the optimal front with solution diversity, serving as a reliable and interpretable measure of algorithm performance in multi-objective optimization.

5.2.2. Assessing the performance of the proposed algorithm using pareto front

Figure 3 illustrates the evolution of Pareto fronts for the optimization algorithm across 1000 generations for sensor configurations with 100, 150, 200, and 250 sensors. The Pareto fronts for all configurations demonstrate rapid convergence in the initial generations (e.g., Gen 5, Gen 10, Gen 50), with significant improvements in reducing total energy consumption as the number of active sensors is optimized. In later generations (e.g., Gen 100, Gen 1000), the Pareto fronts stabilize, showing minimal changes, indicating that the algorithm has reached convergence and identified near-optimal solutions in the Pareto space.

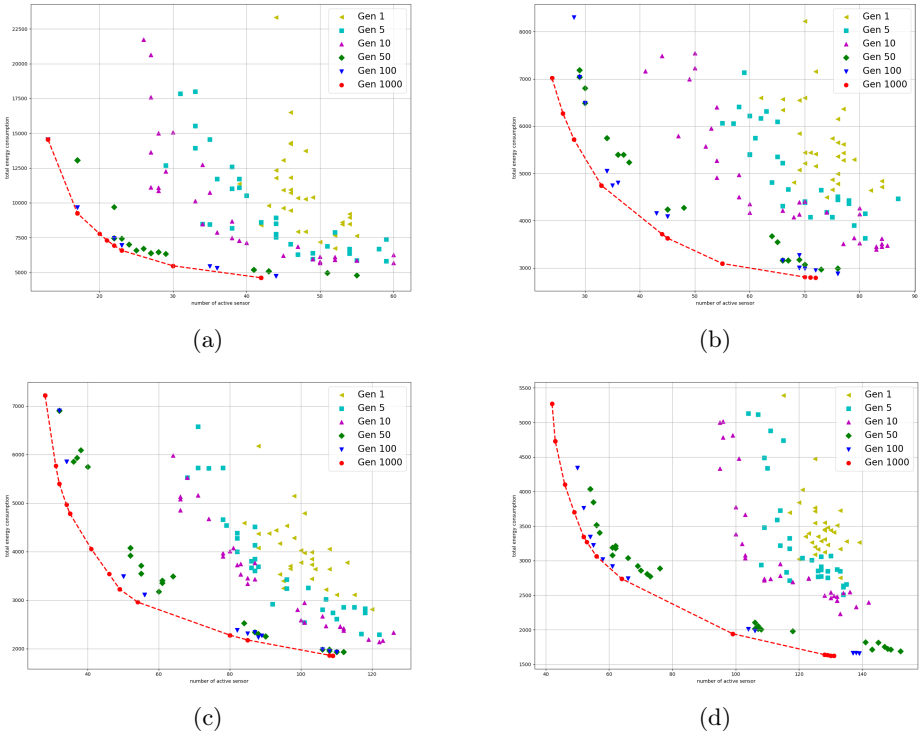


Fig. 3: Illustration of Pareto for different sensor configurations: (a) 100 sensors, (b) 150 sensors, (c) 200 sensors, and (d) 250 sensors.

The algorithm maintains high performance as the problem size scales. It demonstrates the ability to converge and diversify Pareto solutions even with an increasing number of sensors, highlighting its scalability and suitability for solving large-scale multi-objective optimization problems. Notably, the larger configurations (200 and 250 sensors) achieve a broader and more diverse Pareto front compared to smaller configurations, indicating the algorithm's scalability and ability to handle larger search spaces effectively.

Overall, the results demonstrate the optimization algorithm's effectiveness in balancing conflicting objectives for sensor networks of various sizes. The algorithm's rapid initial convergence and the stability of the final Pareto fronts highlight its efficiency and reliability. These findings confirm the algorithm's suitability for real-world applications requiring a balance between energy efficiency and functionality.

6 Conclusion

This paper proposes a weighted-sum approach for optimizing strong barrier coverage in heterogeneous IoT wireless sensor networks (WSNs), balancing two conflicting goals: minimizing active sensors and reducing energy consumption. The flexible model allows network designers to adjust priorities based on specific needs. Experimental results show that this approach effectively reduces active sensors, conserves energy, and extends network lifetime, making it well-suited for resource-constrained IoT environments. Future work could explore adaptive weight adjustments and advanced algorithms to enhance scalability and adaptability for larger, more complex networks, supporting resilient and efficient IoT deployments.

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